

What is a Vector Database?

A vector database indexes and stores vector embeddings for fast retrieval and similarity search, with capabilities like CRUD operations, metadata filtering, horizontal scaling, and serverless.

We're in the midst of the AI revolution. It's upending any industry it touches, promising great innovations - but it also introduces new challenges. Efficient data processing has become more crucial than ever for applications that involve large language models, generative AI, and semantic search.

All of these new applications rely on vector embeddings, a type of vector data representation that carries within it semantic information that's critical for the AI to gain understanding and maintain a long-term memory they can draw upon when executing complex tasks.

Embeddings are generated by AI models (such as Large Language Models) and have many attributes or features, making their representation challenging to manage. In the context of AI and machine learning, these features represent different dimensions of the data that are essential for understanding patterns, relationships, and underlying structures.

That is why we need a specialized database designed specifically for handling this data type. Vector databases like Pinecone fulfill this requirement by offering optimized storage and querying capabilities for embeddings. Vector databases have the capabilities of a traditional database that are absent in standalone vector indexes and the specialization of dealing with vector embeddings, which traditional scalar-based databases lack.

The challenge of working with vector data is that traditional scalar-based databases can't keep up with the complexity and scale of such data, making it difficult to extract insights and perform real-time analysis. That's where vector databases come into play –

they are intentionally designed to handle this type of data and offer the performance, scalability, and flexibility you need to make the most out of your data.

We are seeing the next generation of vector databases introduce more sophisticated architectures to handle the efficient cost and scaling of intelligence. This ability is handled by serverless vector databases, that can separate the cost of storage and compute to enable low-cost knowledge support for AI.

With a vector database, we can add knowledge to our AIs, like semantic information retrieval, long-term memory, and more. The diagram below gives us a better understanding of the role of vector databases in this type of application: